Harsh Kumar Singh

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Education

- **Indian Institute of Information Technology, Lucknow** — Lucknow, India
  Bachelor of Technology in Computer Science and Business; ***CGPA: 8.87/10*** — *Dec 2022 - Jun 2026*
- **Smt. Draupadi Devi Tripathi Intermediate College** — Rudrapur Khajani , Gorakhpur UP
  CBSE Class X - 93.8% ; CBSE Class XII - 95.8% — *Feb 2019 - June 2021*

Experience

- **PayPal** — July 2026 – Present
  Software Engineer — *Bengaluru, India*
  - Developed a reusable, thread-safe **Bigtable connection manager** with client caching and configurable client creation, simplifying Bigtable access and enabling reuse across developers.
  - Optimized an **MSR timezone-resolution query**, reducing bytes billed by **99.88%**, slot consumption by **79.5%**, and execution time by **46.6%** through join and window-logic elimination.
  - Developed **SimileCompareApp**, an automated PDF/CSV report comparison tool for legacy-to-ARGO migrations and regression testing, detecting discrepancies against production baselines and reducing manual validation effort.
  - Led an end-to-end **BigQuery on-demand query batching POC**, implementing dynamic **UNION ALL** batching with partition-prunable predicates, reducing slot consumption by ∼**35%** and data processed by **10%** while preserving request-specific filters.
  - **Tech Stack**: Java, Scala, Python, Apache Spark, Google Cloud Bigtable, Google BigQuery, SQL, Maven, Jenkins, JUnit, ScalaTest, SonarQube
- **PayPal** — June 2025 – Aug 2025
  Software Engineering Intern — *Bangalore, India*
  - Completed end-to-end migration of merchant **Transaction Detail Report (TRR)** from a legacy **Java/Oracle** stack to a modern **Spark/Scala**-based Argo platform, improving scalability, maintainability, and performance.
  - Optimized processing of **28-30 million transactions** to complete within **1.5 hours**, reducing processing time from 4–5 hours in the legacy system (**65% improvement**).
  - Managed go-live deployment for TRR migration and resolved production issues post-launch, ensuring stability and uninterrupted report generation.
  - **Tech Stack**: Scala, Apache Spark, Python, Java, BigQuery, Google Cloud Platform, DataProc, SQL, Git, Jenkins, Agile/Scrum

Projects

- **EduGenius** : GitHub
  - Developed an AI-driven personalized learning web platform to deliver adaptive lessons, curated videos, and interactive quizzes tailored to individual learner needs.
  - Integrated Google Gemini API for dynamic content generation and intelligent video recommendations based on topic relevance and difficulty levels.
  - Enabled offline accessibility by implementing PDF export of selected lessons and quizzes with embedded media and answer keys.
  - **Tech Stack**: Next.js, React, TypeScript, Tailwind CSS, Google Gemini API, Drizzle ORM, PostgreSQL, Streamlit

Skills

- **Languages/Frameworks**: C++, Python, Java, Django, Flask, ReactJS, HTML, CSS, JavaScript
- **Databases**: PostgreSQL, MySQL, BigQuery
- **Cloud and Developer Tools**: Git, GitHub, Postman, Google Cloud Platform
- **Academic Coursework**: Data Structures and Algorithms, DBMS, OOPS, Operating System, Computer Networks
- **AI Engineering**: Claude Code, GitHub Copilot, Google Gemini API, LLM API Integration

Achievements

- Finalist in **Flipkart Grid 5.0** and Selected for **Amazon ML Summer School 2024**
- Solved over **1200+** coding problems across multiple competitive programming platforms.